

Evan Williams

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EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Electrical and Computer Engineering

Expected May 2032

Pittsburgh, PA

Cornell University

Master of Engineering in Computer Science

May 2024

Ithaca, NY

• Advisor: Adrian Sampson

Cornell University

Bachelor of Science in Computer Science, Electrical and Computer Engineering

December 2023

Ithaca, NY

RESEARCH EXPERIENCE

• Stanford University/University of California, Berkeley

December 2025 – Present

Independent Researcher

Stanford, CA

- Developing an interactive synthesis system to schedule a compiler from PyTorch to sparse accelerators [1]
- Advised by Rubens Lacouture, Justin Lubin, and Olivia Hsu

• Cornell Computer Systems Lab

January 2022 – May 2024

Undergraduate Researcher, Capra Research Group

Ithaca, NY

- Designed compiler from PyTorch to FPGA-executable Verilog with parallel banking optimizations [2] [3]
- Implemented accelerated pangenomic graph queries using Dahlia, a hardware generation language

• Purdue Duality Lab

September 2020 – May 2021

Undergraduate Researcher

West Lafayette, IN

- Analyzed domain-specific regular expressions for DFA complexity, portability, and back references
- Implemented ranking scheme to cluster groups of regexes using metrics such as F-Score and mistake rate

PUBLICATIONS

Refereed Workshop Papers

- [1] **Evan Williams**, Justin Lubin, Rubens Lacouture, and Olivia Hsu. *Interactive Compiler Scheduling with Strong Guarantees*. Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE) co-located with ASPLOS. 2026.
- [2] Jiahao Xie, **Evan Williams**, and Adrian Sampson. *From PyTorch to Cayx: An Open-Source Compiler Toolchain for ML Accelerators*. Workshop on Compilers for Machine Learning (C4ML) co-located with CGO. 2026.
- [3] Jiahao Xie, **Evan Williams**, and Adrian Sampson. *From PyTorch to Cayx: An Open-Source Compiler Toolchain for ML Accelerators*. Workshop on Accelerated Machine Learning (AccML) co-located with HiPEAC. 2026.

TEACHING EXPERIENCE

• CS 3410: Computer System Organization and Programming

Course Developer

Summer 2024

Cornell University

• CS 3110: Data Structures and Functional programming

Teaching Assistant

Spring 2022, Spring 2024

Cornell University

• ECE 4750/CS 4420: Computer Architecture

Teaching Assistant

Fall 2023

Cornell University

• ECE 5755: Modern Computer Systems and Architecture

Course Developer

Summer 2023

Cornell Tech

• CS 4820: Intro to Analysis of Algorithms

Teaching Assistant

Fall 2022, Spring 2023

Cornell University

• CS 159: C Programming

Teaching Assistant

Spring 2021

Purdue University

HONORS AND AWARDS

- **Computer and Information Science and Engineering Graduate Fellowship (CSGrad4US)** 2025
National Science Foundation (NSF)
- **Cum Laude** 2023
Cornell University
- **Dean's List** 2021, 2022, 2023
Cornell University
- **Outstanding Undergraduate Course Staff** 2022, 2023, 2024
Cornell University College of Computing and Information Science (CIS)
- **Victor H. and Helen T. Green ECE Scholarship** 2021
Purdue University School of Electrical and Computer Engineering (ECE)

WORK EXPERIENCE

- **Amazon Web Services** August 2024 – April 2026
Software Development Engineer, Amazon HealthOmics Mountain View, CA
 - Led effort to support CUDA driver integration and implemented type coercion to enable new customer workloads
 - Parallelized integration test execution to reduce time for production deployments in CI/CD pipeline by 50%
- **Amazon Web Services** May 2023 – August 2023
Software Development Engineer Intern, AWS IoT RoboRunner Sunnyvale, CA
 - Developed application to perform deep packet inspection on MQTT network traffic using C++ and Docker
 - Integrated with Amazon CloudWatch and AWS DeviceDefender to alert customers of anomalous behavior
- **Deloitte Consulting** June 2022 – August 2022
Solutions Engineer Intern New York, NY
 - Conducted research and drafted design documentation for mobile application to assist crews in electrical outages
 - Composed Python scripts to automate SQL database queries and connected to live front-end using a REST API
- **Commonwealth Associates** May 2021 – August 2021
Electrical Engineering Intern Jackson, MI
 - Designed alarm modules and relay circuits for electrical substations to improve power and energy efficiency
 - Created mobile application for internal employees to express feedback and ideas to upper-level management

TALKS

- **Compiler Infrastructures for Hardware Accelerator Design** April 2026
Google
- **Frontend Accelerators for Pangenomic Graph Queries** May 2022
NSF PPoS Panorama Meeting for Computational Genomics